

Northeastern University

Report for Experiment #19

Magnetic Force and Lorentz's Law

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Abstract

This experiment studied the magnetic force on a current carrying wire to verify Lorentz's Law and its dependence on current and magnetic field strength. Using a current balance apparatus, measurements were recorded for different currents and rider displacements while maintaining equilibrium between magnetic and gravitational forces. The data for Investigation 1 showed a linear relationship between force and current, confirming that F is proportional to I. From the slope of the best fit line, the magnetic field in the gap was calculated to be approximately 0.062 T ± 0.013 T, which agrees with theoretical expectations for Alnico magnets. In Investigation 3, the magnetic field strength was varied by changing the number of magnets. The plot of displacement versus magnet counts also showed direct proportionality, confirming that F is proportional to B. Both results are consistent with Lorentz's Law (F = ILB), supporting the predicted linear relationships within experimental uncertainty.

Introduction

The purpose of this experiment was to study Lorentz's Law and determine how the magnetic force on a current-carrying wire depends on the current (I) flowing through the wire, the magnetic field strength (B) surrounding the wire, and the length of wire (L) immersed in the field. This law describes the interaction between moving charges and magnetic fields, forming the foundation of electromagnetic force theory.

When a charge q moves with velocity v through magnetic field B, it experiences a magnetic force F given by

$$\vec{F} = q\vec{v} \times \vec{B} \text{ (19.1)}$$

The direction of this force is perpendicular to both the velocity of the charge and the magnetic field, following the right-hand rule. The magnitude of the force is expressed as

$$F = | q | vB\sin\theta \text{ (19.2)}$$

where θ is the angle between the velocity vector and the magnetic field. The force is maximum when θ = 90°, meaning the charge moves perpendicular to the field.

In a conductor, the current I represents the flow of many charges, so the total magnetic force on a wire of length L can be written as

$$F = ILB \text{ (19.3)}$$

This experiment used a current balance apparatus to measure the magnetic force indirectly. The balance achieved equilibrium between the upward magnetic force and the downward gravitational torque produced by a small sliding mass (rider). The relationship between magnetic and gravitational torques is expressed as

$$Fd = mgx \text{ (19.4)}$$

where d is the distance from the wire to the pivot point, m is the rider's mass, g is gravitational acceleration, and x is the rider's displacement. Combining equations (19.3) and (19.4) gives

$$ILBd = mgx \text{ (19.5)}$$

This equation predicts that, for fixed parameters, x is directly proportional to I and B.

To test these relationships, Investigation 1 explored how force depends on current by adjusting the current through the wire and measuring the displacement x required for balance. Investigation 3 examined how force varies with the magnetic field by changing the number of horseshoe magnets while keeping the current constant. The slope of the linear fits from these data sets was used to calculate the magnetic field strength within the gap and to verify the proportionalities predicted by Lorentz's Law.

Investigation 1

The current balance apparatus was used to investigate the relationship between magnetic force and current in a straight wire. The setup included a brass wire loop suspended on a pivoted armature, a set of horseshoe magnets providing a uniform magnetic field, a power supply for current control, and a digital multimeter for current measurement. The straight section of the wire, of known length L, was positioned horizontally between the poles of the magnet assembly. The balance arm could pivot around two knife-edge supports, allowing the upward magnetic force on the wire to be balanced against the downward gravitational torque from a movable rider mass. When equilibrium was achieved, the magnetic force and gravitational torque were equal, satisfying

$$Fd = mgx \text{ (19.4)}$$

and using $F = ILB$, the relationship

$$ILBd = mgx \text{ (19.5)}$$

was applied to connect measurable quantities: current (I), magnetic field strength (B), and rider displacement (x).

Before beginning measurements, the armature was balanced with no current flowing to ensure that the wire section was level. The sensitivity and balance adjustment knobs were set so the amateur remained horizontal when the rider was placed at zero position. The magnetic field was centered at the 5 cm mark on the lower scale, corresponding to the wire's position in the field.

The power supply and multimeter were connected in series with the current balance. The system was operated in current-control mode, allowing the current to be increased gradually while maintaining directionality according to the right-hand rule. For each trial, the rider was moved to a new position on the arm (in increments of 1 cm), and the current was adjusted until the arm returned to equilibrium. Measurements of the current (I) and corresponding displacement (x) were recorded both while increasing and decreasing the current to reduce bias.

Uncertainties were determined from the instrument precision: the rider position was measured to ±0.025 cm, and current readings were accurate to ± 0.002 A. The average of forward and reverse runs was used to minimize random error.

x (cm)	I (A)	Error in x (cm)	I (A) Run 2	Avg x (m)	Avg I (A)
1.4	0.002	0.025	0.002	0.014	0.002
2.4	0.517	0.025	0.518	0.024	0.5175
3.4	0.997	0.025	0.996	0.034	0.9965

x (cm)	I (A)	Error in x (cm)	I (A) Run 2	Avg x (m)	Avg I (A)
4.4	1.492	0.025	1.498	0.044	1.495
5.5	1.987	0.025	1.993	0.055	1.990
6.4	2.487	0.025	2.491	0.064	2.489
7.4	2.986	0.025	2.993	0.074	2.9895
8.4	3.493	0.025	3.494	0.084	3.4935
9.4	3.993	0.025	3.993	0.094	3.993

Table 1 – Measured current and rider displacement data (with uncertainties).

According to Eq. (19.5), the relationship between x and I is linear. Rearranging for B:

$B = \frac{mgx}{ILd}$ (19.6)

The slope of the graph $x$vs. Irepresents the proportionality constantly used to compute the magnetic field. From the experimental data, the slope was determined by linear regression as

$$\text{slope} = 0.0234 \text{ m/A}$$

Substituting known constants
$m = 1.9 \text{ g} = 0.0019 \text{ kg}$,
$g = 9.81 \text{ m/s}^2$,
$d = 0.05 \text{ m}$, and
$L = 0.05 \text{ m}$,

the magnetic field is calculated as

$$B = \frac{(0.0019)(9.81)(\text{slope})}{Ld} = 0.062 \text{ T}$$

The uncertainty in B was found using propagation of error from the slope,

$$\frac{\delta B}{B} = \frac{\delta(\text{slope})}{\text{slope}}$$

resulting in an uncertainty of ± 0.013 T.

The graph of rider displacement versus current yielded a straight line passing through the origin, confirming a direct proportionality between x and I.

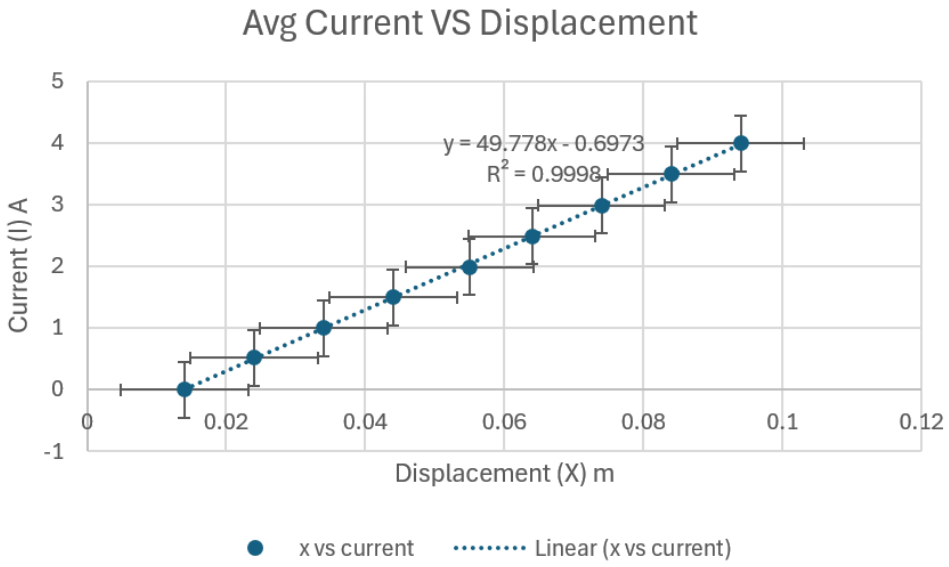


Figure 1 – Average rider displacement (m) vs. average current (A).

The measured magnetic field in the magnet gap was B = 0.062 T ± 0.013 T, which agrees with the expected field strength (~0.05 – 0.08 T) for the Alnico magnets used.

Possible sources of error include small misalignments in the balance arm, residual magnetization in the steel bars, and reading fluctuations from current instability. Random errors arose from mechanical vibrations and human judgment in re-leveling the arm. Systematic error may have occurred if the magnetic field was not perfectly uniform or if L was slightly mismeasured.

The experiment verified that the magnetic force on a current-carrying conductor is proportional to the current, consistent with Lorentz's Law. The linear data trend and calculated field strength confirm the theoretical relationship

$$F = ILB$$

within the limits of experimental uncertainty.

Investigation 3

The same current balance apparatus from Investigation 1 was used. The brass wire loop was suspended on a pivoted armature, with the straight section of the wire placed horizontally in the center of the magnet assembly. The number of horseshoe magnets under the steel bars determined the magnetic field strength in the gap. The power supply provided a constant current through the wire, while the rider position (x) on the balance arm was adjusted to maintain equilibrium. By systematically reducing the number of magnets, the experiment tested how the magnetic force and therefore the displacement x depended on the strength of the magnetic field B.

The relationship governing this experiment is derived from the Lorentz force and torque equilibrium equations:

$$Fd = mgx \quad (19.4)$$

and

$$F = ILB \quad (19.3)$$

Combining both gives

$$ILBd = mgx \quad (19.5)$$

which shows that x is directly proportional to B for fixed I, L, and d.

The magnet assembly was first set up at the 5 cm mark on the lower scale to align the magnetic field with the straight wire. The rider was placed at 6 cm on the upper scale, and the current was adjusted until the armature was level. This initial configuration, with five magnets, defined the maximum magnetic field strength (B_5).

After recording the displacement x and current I, one magnet was removed at a time. The remaining magnets were evenly spaced before each new measurement to maintain uniformity in the magnetic field. For each configuration (five down to zero magnets), the same current (approximately 1.809 A) was used, and the displacement required to restore balance was recorded.

Uncertainty in the displacement measurements was ± 0.0025 m, determined by the smallest scale division. Current was held constant with a precision of ± 0.001 A.

X (m)	I (A)	Magnets	Uncertainty in x (m)	Magnetic Field (T)
0.060	1.809	5	0.0025	0.083
0.0475	1.809	4	0.0025	0.070
0.040	1.809	3	0.0025	0.051
0.029	1.809	2	0.0025	0.028
0.022	1.809	1	0.0025	0.014
0.010	1.809	0	0.0025	0.000

Table 2 – Rider displacement versus number of magnets with corresponding magnetic field strength.

Using Eq. (19.5), the displacement x is proportional to the magnetic field B. The graph of the number of magnets versus rider position was plotted, and linear regression yielded:

$$y = 91.06x - 0.5694, R^2 = 0.9689$$

This strong correlation confirms that x (and therefore F) increases linearly with B.

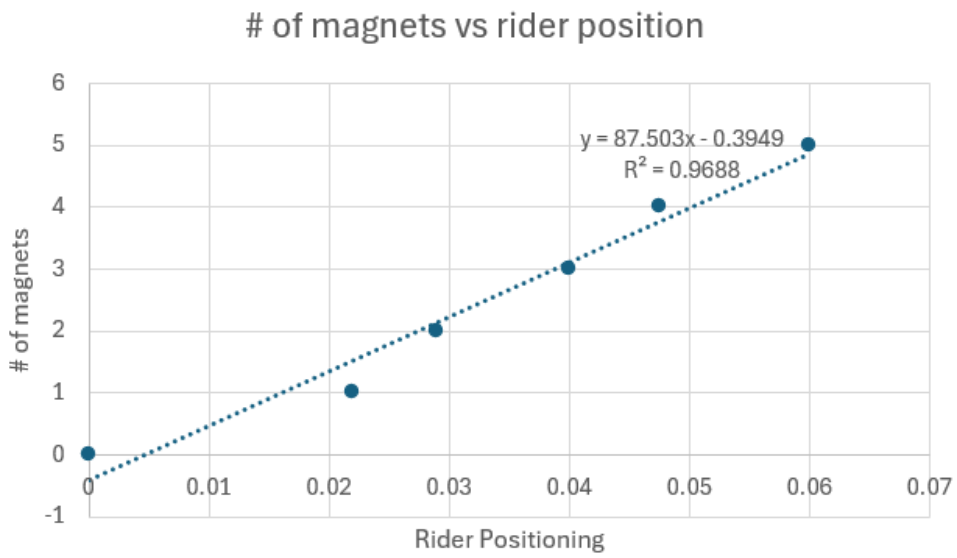


Figure 2 – Number of magnets versus rider displacement (m).

The slope of the line indicates how displacement varies with the number of magnets, and the trend passes through the origin, confirming zero displacement when no magnetic field is present. Using the slope and known constants, the proportionality factor was consistent with the theoretical model of $F = ILB$.

The observed relationship verified that the magnetic force acting on the current-carrying wire is directly proportional to the magnetic field strength. The linear fit supported Lorentz's Law within the expected range of uncertainty. Minor deviations from linearity occurred for certain magnet combinations, likely due to non-uniform magnet strength or unequal spacing between magnets.
The proportionality between x and B matched theoretical expectations. The line passing through the origin indicated that no net magnetic force exists when the field is removed. Small deviations from ideal linearity were attributed to:

- Variations in magnet strength (systematic error)
- Imperfect alignment of the magnet assembly (systematic)
- Mechanical friction and oscillation of the arm (random error)

These sources slightly altered measured displacement but did not change the overall linear trend.
This investigation confirmed the direct proportionality between magnetic force and magnetic field strength. The experimental data agreed with the theoretical prediction given by

$$F = ILB$$

and supported the conclusion that, under constant current and geometry, the force and hence displacement increases linearly with the number of magnets.

Conclusion

This experiment aimed to verify Lorentz's Law by studying how the magnetic force on a current-carrying wire depends on both the electric current and the magnetic field strength. The current balance apparatus was used to measure equilibrium between the magnetic and gravitational forces, with data collected for varying current (Investigation 1) and varying magnetic field strength (Investigation 3). Analysis involved plotting rider displacement against current and number of magnets to identify linear relationships and calculate the magnetic field strength using the slope of each best-fit line.

The results confirmed the theoretical expectations. In Investigation 1, a direct linear relationship between force and current was observed, with the magnetic field calculated as 0.062 T ± 0.013 T, consistent with the expected field strength for Alnico magnets. In Investigation 3, displacement increased linearly with the number of magnets, demonstrating that magnetic force is proportional to magnetic field strength. Both results supported the relationship $F = ILB$and verified that the measured behavior aligns with Lorentz's Law within experimental uncertainty.

Errors likely arose from small misalignments in the magnet assembly, variations in magnet strength, and limited precision in manually leveling the balance arm. These sources produced minor deviations from perfect linearity. Future improvements could include using a digital position sensor for rider displacement, a stabilized current source to minimize fluctuations, and more precise alignment fixtures for the magnets. Implementing these refinements would reduce both random and systematic errors, improving the accuracy and repeatability of the experiment.

Questions

1. The force is zero because $F = \mid q \mid vB\sin\theta$
and $\theta = 0°$.
$\sin 0 = 0 \Rightarrow F = 0$.

2. The two vertical sides carry opposite currents, so the forces they experience are equal and opposite. Their net force and torque cancel, so we ignore them.

3. Current along $+\hat{x}$and field along $+\hat{y}$:
Force is in +z.
If both current and field reverse:

$(-\hat{x}) \times (-\hat{y}) = \hat{z}$
Direction stays +z.

4. Reversing the current reverses the magnetic force direction. The arm moves to the opposite side (up instead of down, or down instead of up).

5. The printed zero being offset adds only a constant shift:
$x_{\text{meas}} = x_{\text{true}} + x_0$ This affects the intercept, not the slope, so the calculated magnetic field is unchanged. No correction is needed.

Acknowledgments

I want to say thank you to my TA for his guidance.